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CASE STUDY REPORT

Submitted
By

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CASE STUDY 1

Nmap Installation and Network Scanning

1. Introduction

Securing a computer starts with knowing what it exposes to the network. Nmap (Network Mapper) is a free and open-source tool for network discovery and security auditing. It sends crafted probes to a target and classifies each port as open, closed or filtered, and can optionally identify the service and its version running behind an open port.

In this case study, only the student's own machine (localhost) was scanned.

2. Objectives

- To install Nmap from its official distribution source and verify the installation.
- To identify the test machine's address and scan only localhost, an authorized target.
- To perform a basic port scan and record the observed results.
- To explain the meaning of open, closed and filtered ports.
- To recommend one security improvement based on an observed result.

3. Problem Statement

A laboratory administrator wants to document which network services are listening on a test computer and determine whether unnecessary services should be disabled. The task is to use Nmap, within authorized limits, to discover the listening services, interpret the port states and recommend a hardening step.

4. Implementation and Procedure (Steps, Commands, Tools and Software Used)

4.1 Environment, Tools and Software Used

Item	Details
Operating system	Microsoft Windows, version 10.0.26300.9711 (as reported by Command Prompt)
Tool	Nmap 7.992 (command line) with Npcap, from nmap.org (official source)
Command interface	Command Prompt / Windows PowerShell
Scope / authorization	Own device only. Target: 127.0.0.1 (localhost). No other host or network was scanned.

4.2 Steps and Commands

Step 1: Install Nmap from the official source and verify it

The Windows installer (nmap-setup.exe) was downloaded from the official download page, https://nmap.org/download.html, and run with the default options. The installer also installs Npcap, the packet capture driver Nmap uses on Windows. Installation was verified by checking the version:

```
nmap --version
```

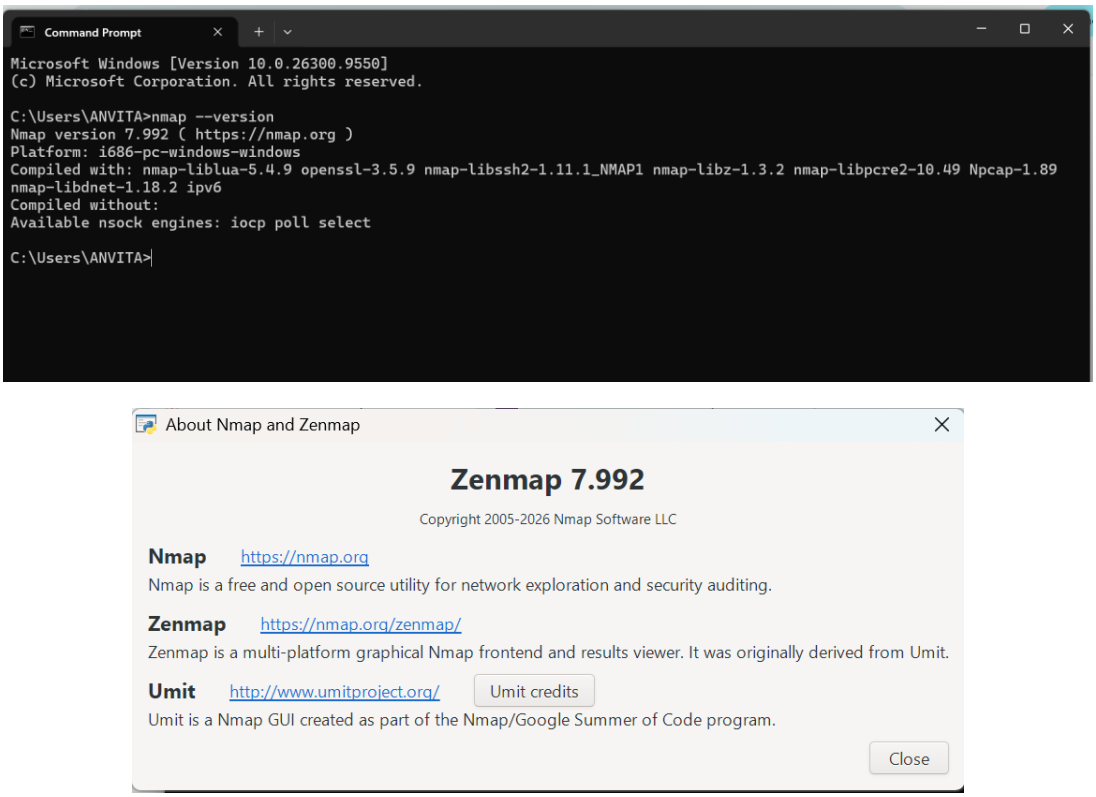


Figure 1.1: Nmap installed and version verified

Step 2: Identify the test machine's address and select the target

The machine's own address was found with `ipconfig`. Only the loopback address 127.0.0.1 (localhost), which refers to the same computer, was used as the scan target, because it is a system the student owns and is authorized to scan.

`Ipconfig`

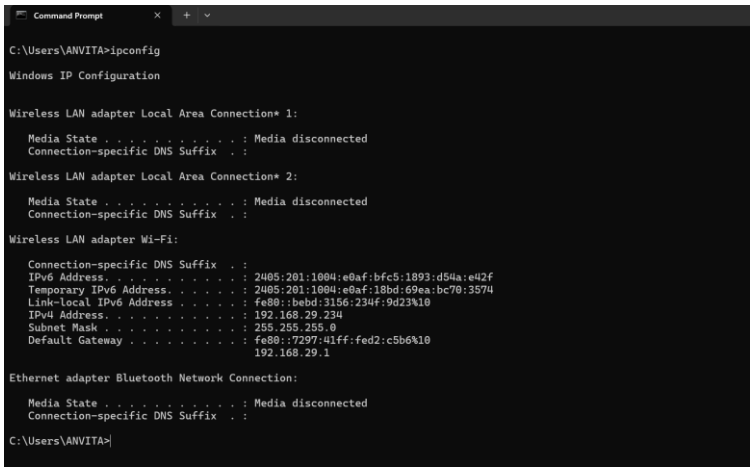


Figure 1.2: Test machine address (ipconfig) and chosen target (localhost)

Step 3: Perform a basic port scan

A TCP connect scan of the localhost ports was performed, first with the default port list and then across all 65,535 ports with service version detection:

```
nmap -sT 127.0.0.1
nmap -sT -sV -p- 127.0.0.1
```

Option	Meaning
-sT	TCP connect scan: completes the normal TCP handshake; works without raw-socket access, so it is reliable on Windows localhost.
-sV	Service/version detection: probes open ports to identify the service and version.
-p-	Scan all 65,535 TCP ports instead of the default top 1000.
127.0.0.1	Target: the local machine (loopback address).

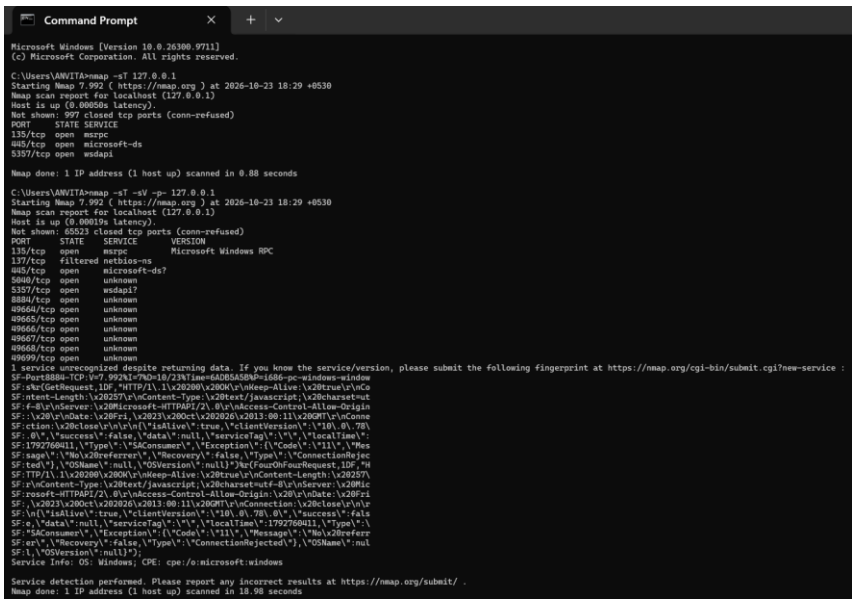


Figure 1.3(a): Nmap scan output for 127.0.0.1 on command prompt

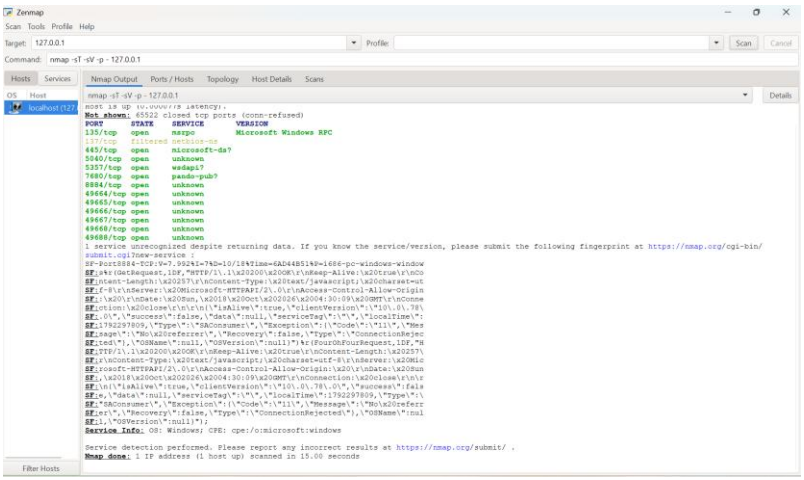


Figure 1.3(b): Nmap scan output for 127.0.0.1 on ZenMap

Step 4: Interpret the results and examine one open port

The output was read line by line (PORT, STATE, SERVICE, VERSION). One open port was then selected and probed on its own to document it in detail:

```
nmap -sT -sV -p 445 127.0.0.1
```

```
C:\Users\ANVITA>nmap -sT -sV -p 445 127.0.0.1
Starting Nmap 7.992 ( https://nmap.org ) at 2026-10-23 18:32 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00s latency).

PORT    STATE SERVICE       VERSION
445/tcp open  microsoft-ds?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.59 seconds

C:\Users\ANVITA>
```

Figure 1.4: Detailed scan of one observed port

Step 5: Recommend a security improvement

Based on the observed open port, one hardening recommendation was prepared.

5. Results and Output

Two scans of 127.0.0.1 were run (Figure 1.3). The default scan checks the 1000 most common TCP ports; the full scan checks all 65,535 ports with service version detection. The default scan found **3 open ports** (135, 445 and 5357) and 997 closed ports, and finished in 0.88 seconds. The full scan found **11 open ports, 1 filtered port and 65,523 closed ports** (connection refused), and finished in 18.98 seconds. Nmap identified the operating system family as Windows.

Port	State	Service	Version / notes
135/tcp	open	msrpc	Microsoft Windows RPC (RPC endpoint mapper)
137/tcp	filtered	netbios-ns	NetBIOS name service; probes were dropped (no reply)
445/tcp	open	microsoft-ds?	Windows file sharing (SMB); "?" means Nmap guessed the service from the port number
5040/tcp	open	unknown	Not identified by Nmap
5357/tcp	open	wsdapi?	Web Services for Devices (device discovery); guessed from the port number
8884/tcp	open	unknown	Replied to an HTTP request with data from a Microsoft HTTP API (Microsoft-HTTPAPI/2.0) server, but Nmap could not name the service
49664 to 49668/tcp, 49699/tcp	open	unknown	High-numbered ports in the dynamic range Windows services use for RPC; six ports in total

A third scan targeted only port 445 (`nmap -sT -sV -p 445 127.0.0.1`, Figure 1.4). It reported the port as open, service "microsoft-ds?", and finished in 6.59 seconds.

6. Applications and Security Analysis

Meaning of open, closed and filtered ports

State	Meaning	Security implication
Open	A service is actively listening and accepts connections (the TCP handshake completes).	Part of the attack surface. It must be needed, patched and restricted.
Closed	The port is reachable, but no application is listening; the host replies that the connection is refused (TCP RST).	No service to attack, but it confirms the host is alive.
Filtered	Nmap cannot tell whether the port is open or closed because a firewall or filter drops the probes or answers with an error.	Suggests a firewall is hiding the port. Reveals less to an observer.

Explanation of one observed port

Taking **port 445/tcp (microsoft-ds)** as the example, this port is used by the Server Message Block (SMB) protocol for Windows file and printer sharing. The scan showed it as open, meaning the Windows Server service is listening and willing to accept connections. This is normal on a machine that shares files, but it is risky when sharing is not needed, because SMB has been the target of serious worm attacks in the past (for example WannaCry, which abused a flaw in the older SMBv1 protocol).

Recommended security improvement

If the laboratory computer does not need to share files, the SMB service should not be reachable. The recommended improvement is to turn off File and Printer Sharing for the active network profile (or block inbound TCP 445 in the host firewall), and to confirm that the obsolete SMBv1 protocol is disabled. After the change, the scan should be repeated; port 445 should then appear as closed or filtered. The same approach applies to any unnecessary open port: identify the service, decide whether it is required, then disable it or restrict it to trusted addresses.

Applications of port scanning

- Building an inventory of listening services on servers and lab machines.
- Verifying that firewall rules and hardening steps actually work.
- Security audits and the reconnaissance stage of authorized penetration tests.
- Troubleshooting, for example checking whether a web or database service is reachable.

Because the same technique is used by attackers, scanning must be limited to systems owned by the tester or explicitly authorized in writing; unauthorized scanning may violate law and institutional policy. A limitation of this exercise is that scanning localhost only shows what is listening on the machine itself; it does not show how the machine appears to other computers across the network firewall.

7. Conclusion

Nmap was installed from its official source and used to scan only the student's own machine. The scan identified the open ports, and the difference between open, closed and filtered states was explained. An open port that is not required, such as SMB on port 445, increases the attack surface, and the recommended fix is to disable or restrict the service and re-scan to confirm the change. All activities were performed on the student's own device, within the authorization limits stated in the assignment.

8. References

[1] Nmap Project. Nmap download page (official distribution). https://nmap.org/download.html

[2] Nmap Project. Nmap Reference Guide (man page). https://nmap.org/book/man.html

[3] Nmap Project. Port Scanning Basics (port states). https://nmap.org/book/man-port-scanning-basics.html

[4] NIST SP 800-115. Technical Guide to Information Security Testing and Assessment.

CASE STUDY 2

Antivirus Configuration and Security Review

1. Introduction

Antivirus or endpoint protection software is the host-level control that detects and blocks known malicious files using signatures, heuristics, behaviour monitoring and cloud intelligence. Its value depends on three things: it must be active, its detection database must be current, and its settings must be sensible.

This case study reviews those three points on a laboratory computer and confirms detection safely using the industry-standard EICAR test file, which is harmless but is recognised by antivirus products as a test threat. No real malware was downloaded, created or executed.

2. Objectives

- To identify the antivirus product installed on the system.
- To inspect its protection status, security intelligence updates and relevant settings.
- To run an approved quick scan and review the result.
- To demonstrate detection safely using the EICAR test file, without using real malware.
- To explain the difference between antivirus detection and a guarantee that a system is free from all threats.

3. Problem Statement

A laboratory user needs to verify that endpoint protection is active, updated and configured to detect suspicious files. The task is to inspect the antivirus status and settings, run an approved scan, demonstrate detection with the EICAR test file, and explain what antivirus can and cannot guarantee.

4. Implementation and Procedure (Steps, Commands, Tools and Software Used)

4.1 Environment, Tools and Software Used

Item	Details
Operating system	Microsoft Windows, version 10.0.26300.9711
Antivirus (active)	McAfee with McAfee WebAdvisor browser extension
Other product registered	Windows Defender (registered with Windows Security alongside McAfee)
Command interface	Windows PowerShell (Administrator), used only to list the installed security products
Test file	EICAR standard antivirus test file (eicar.com.txt) from eicar.org (official source)
Restrictions followed	No real malware used. Real-time protection was not disabled at any point.

4.2 Steps and Commands

Step 1: Identify the antivirus product and check its protection status

The security products registered with Windows were listed from PowerShell. The output shows both Windows Defender and McAfee. The McAfee application was then opened to read its protection status, and Windows Security (Virus & threat protection) was checked to confirm which product is the active antivirus.

```
Get-CimInstance -Namespace root/SecurityCenter2 -ClassName AntiVirusProduct
                        | Select-Object displayName
```

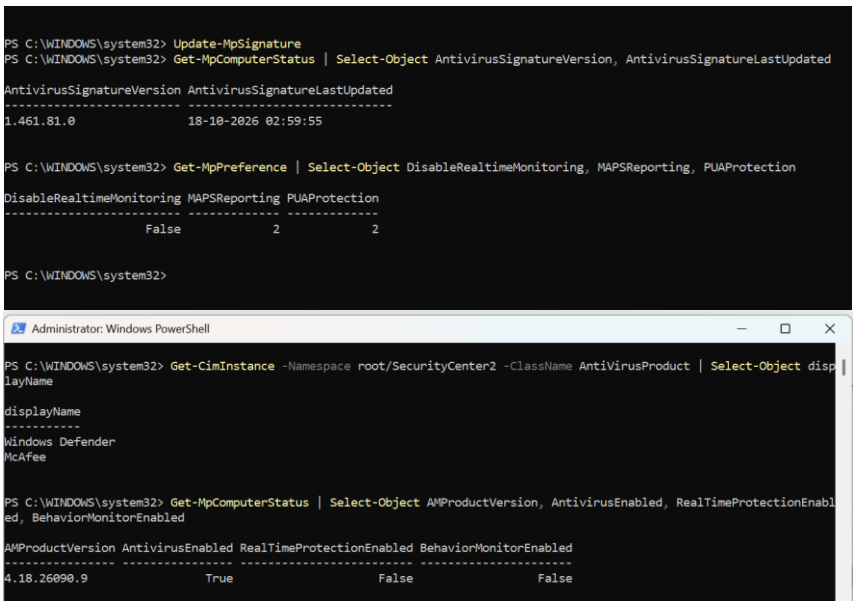


Figure 2.1(a): Protection status on Powershell

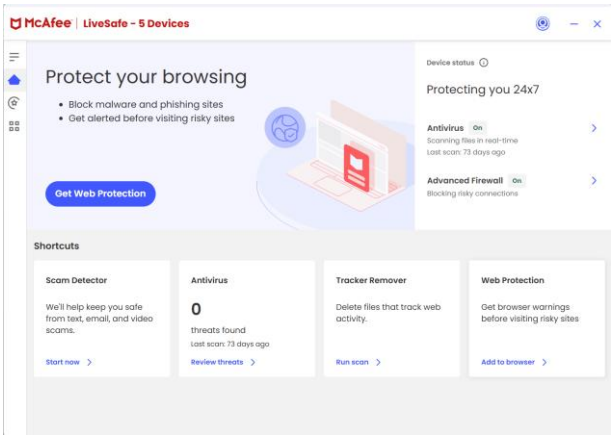


Figure 2.1: Antivirus product (McAfee)

Step 2: Inspect security intelligence updates and settings

In the McAfee application, the update status and the protection settings were reviewed: real-time scanning, automatic updates and web protection. A manual update check was run from the application.

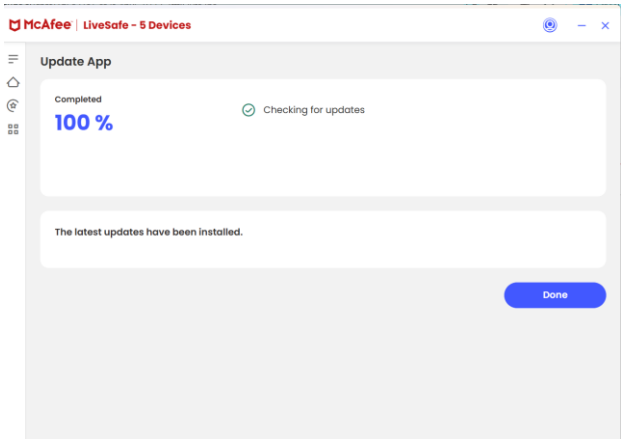


Figure 2.2: Antivirus update status

Step 3: Run an approved quick scan

A quick scan was started from the McAfee application (My Protection > Scans > Quick Scan) and allowed to complete. The result was read from the scan summary screen.

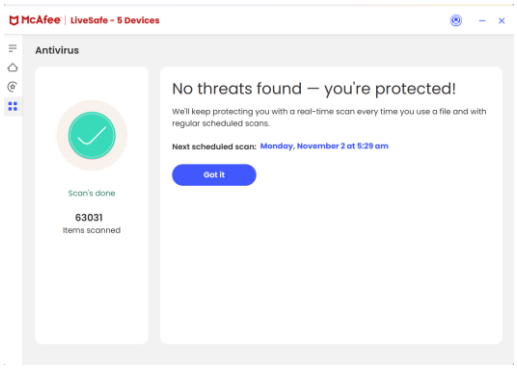


Figure 2.3: Quick scan completed

Step 4: Demonstrate detection with the EICAR test file

The EICAR standard antivirus test file was requested from its official source, https://www.eicar.org/download-anti-malware-testfile/. It is a harmless text file that antivirus products are designed to flag as a test threat. McAfee WebAdvisor first blocked the request, marking the page as a malicious site. The page was then opened deliberately, since it is the official EICAR site and the file is harmless, and the detection of the file itself by McAfee was captured from its alert and quarantine or scan history. The test file was afterwards removed.

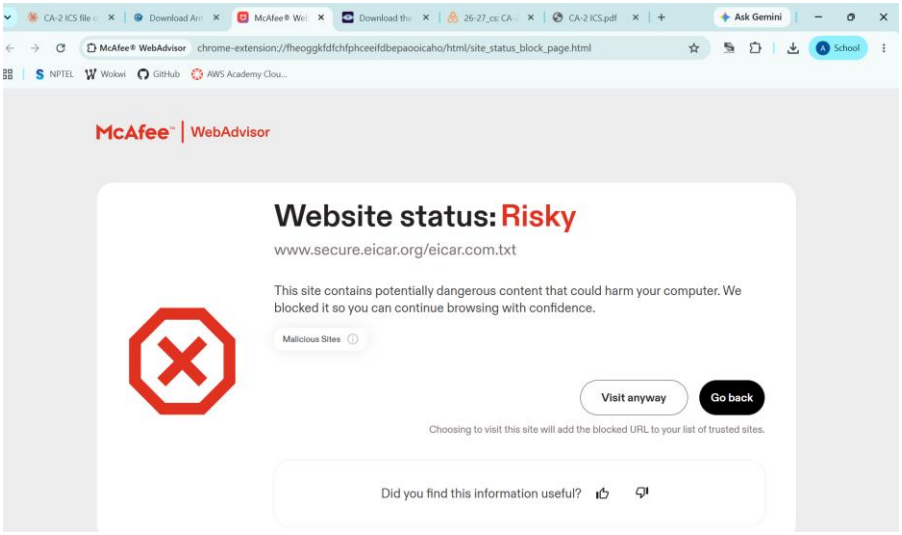


Figure 2.4: EICAR download blocked by McAfee WebAdvisor (web-level detection)

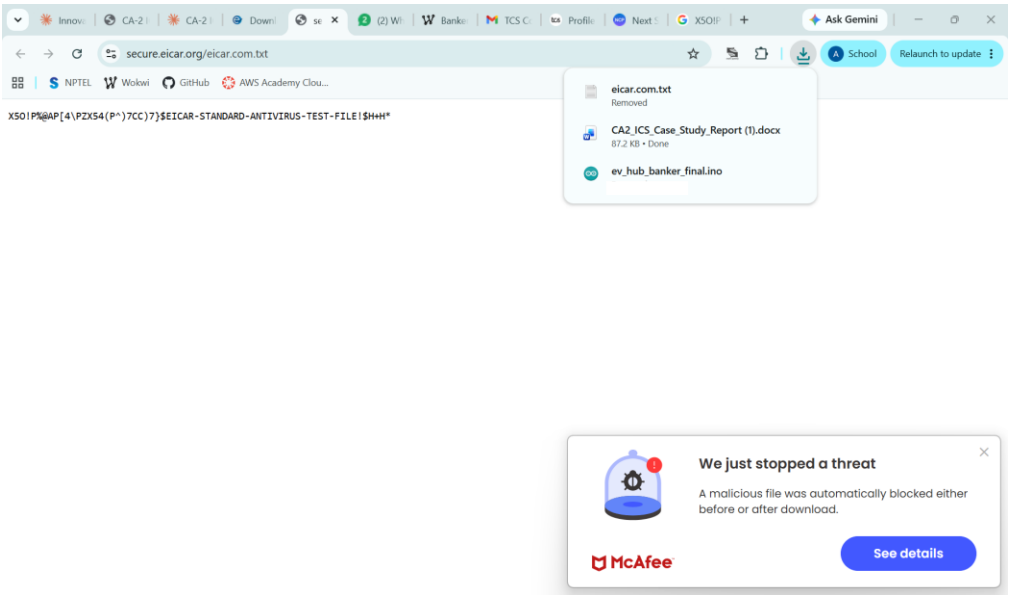


Figure 2.5: EICAR test file detected by McAfee (file-level detection)

Step 5: Explain the limitation

The difference between detection and a guarantee of a threat-free system was documented.

5. Results and Output

Check	Observed result
Active antivirus product	McAfee
Real-time protection	On, as shown in the McAfee app
Virus definition status	Up to date
Quick scan	Completed, 0 threats found
Web protection (EICAR page)	McAfee WebAdvisor blocked www.secure.eicar.org/eicar.com.txt as "Risky / Malicious Sites"
EICAR file detection	Detected and quarantined. The entry is visible in McAfee quarantine / scan history

McAfee was active, its definitions were current, and the EICAR test material was stopped at two levels: the web filter blocked the download page, and the antivirus detected the file itself. This confirms the detection pipeline works as expected.

6. Applications and Security Analysis

Antivirus detection versus a guarantee of a threat-free system

Antivirus detection means the product recognised a file or behaviour as malicious at the moment it was examined, using known signatures, heuristics, behaviour monitoring or cloud lookups. It is not a guarantee that the system is free of all threats, for several reasons:

- **Unknown threats:** brand-new (zero-day) or heavily modified malware may not match any signature yet.
- **Fileless and living-off-the-land attacks:** attacks that use legitimate tools such as scripts leave no obvious malicious file to scan.
- **Non-malware risks:** phishing, stolen passwords, weak configuration and unpatched software are not detected by file scanning.
- **Timing and configuration:** outdated definitions, disabled protection or excluded folders create gaps.

The EICAR test proves only that the detection path is working for a known test pattern. It says nothing about how the product handles unknown malware. Antivirus should therefore be one layer of defence in depth, together with timely patching, a host firewall, least-privilege accounts, regular backups and user awareness.

Applications

- Routine endpoint health checks in labs, offices and enterprises.
- Compliance audits that require evidence of active, updated protection.

- Safe testing of detection and alerting workflows using EICAR rather than real malware.

7. Conclusion

The antivirus product was identified, confirmed to be active and up to date, and verified using a quick scan and the safe EICAR test file. The exercise also showed that antivirus detection is valuable but not a guarantee of safety, so it must be combined with other security controls. No real malware was used and real-time protection stayed on throughout.

8. References

[1] EICAR. Anti-malware test file. https://www.eicar.org/download-anti-malware-testfile/

[2] McAfee. Product support and documentation. https://www.mcafee.com/support

[3] NIST SP 800-83 Rev. 1. Guide to Malware Incident Prevention and Handling for Desktops and Laptops.